

Zomato Restaurant Market Analysis

Bangalore – Exploratory Data Analysis Report

Prepared by: Viraj More | May 2026

virajmore.data888@gmail.com

Tools: Python · Pandas · Matplotlib · Seaborn · Termux + Acode

Dataset: 51,148 Restaurants | Bangalore | Kaggle

Dataset Overview

51,717 → 51,148

Rows (Raw → Cleaned)

17 → 15

Columns (Raw → Cleaned)

38,045 → 0

Null Values Eliminated

Metric	Raw Dataset	Cleaned Dataset
Total Rows	51,717	51,148
Total Columns	17	15
Total Nulls	38,045	0
Duplicates	0	0
Rows Dropped	—	569
Columns Dropped	—	phone, dish_liked

Key EDA Metrics – At a Glance

51,148

Restaurants Analysed

₹400

Median Cost for Two

3.68 / 5

Average Rating

1,945

Value-for-Money Picks

14.57M

Total Votes

Rating Distribution

Metric	Value
Min	1.8
25th %ile	3.4
Median	3.7
Mean	3.68
75th %ile	3.9
Max	4.9
Std Dev	0.41

Cost Distribution (For Two)

Metric	Value
Min	₹40
25th %ile	₹300
Median	₹400
Mean	₹556
75th %ile	₹700
Max	₹6,000
Std Dev	₹440

Votes Distribution

Metric	Value
Min	0
25th %ile	7
Median	41
Mean	285
75th %ile	199
Max	16,832
Zero-vote Restaurants	9,863 (19.3%)

Data Cleaning – Steps & Decisions

Every decision was evidence-based — not assumption-based.

1

drop_duplicates()

0 duplicates found — confirmed clean entry

2

Drop 'dish_liked' column

28,078 nulls (54.3% missing) — unusable

3

Drop 'phone' column

1,208 nulls + irrelevant for analysis

4

dropna() on critical cols

location, rest_type, cuisines, cost — 569 rows dropped

5

rate: remove '/5' suffix

Raw format was '4.2/5' — extracted numeric value

6

rate: handle 'NEW' & '-'

2,207 'NEW' + 69 '-' converted to NaN for imputation

7

rate: groupby median fill

Filled NaN using median per rest_type + location group

8

cost: remove commas

'1,000' → '1000' — string cleaning before conversion

9

Downcast dtypes

float16 for rate, int32 for votes — memory optimisation

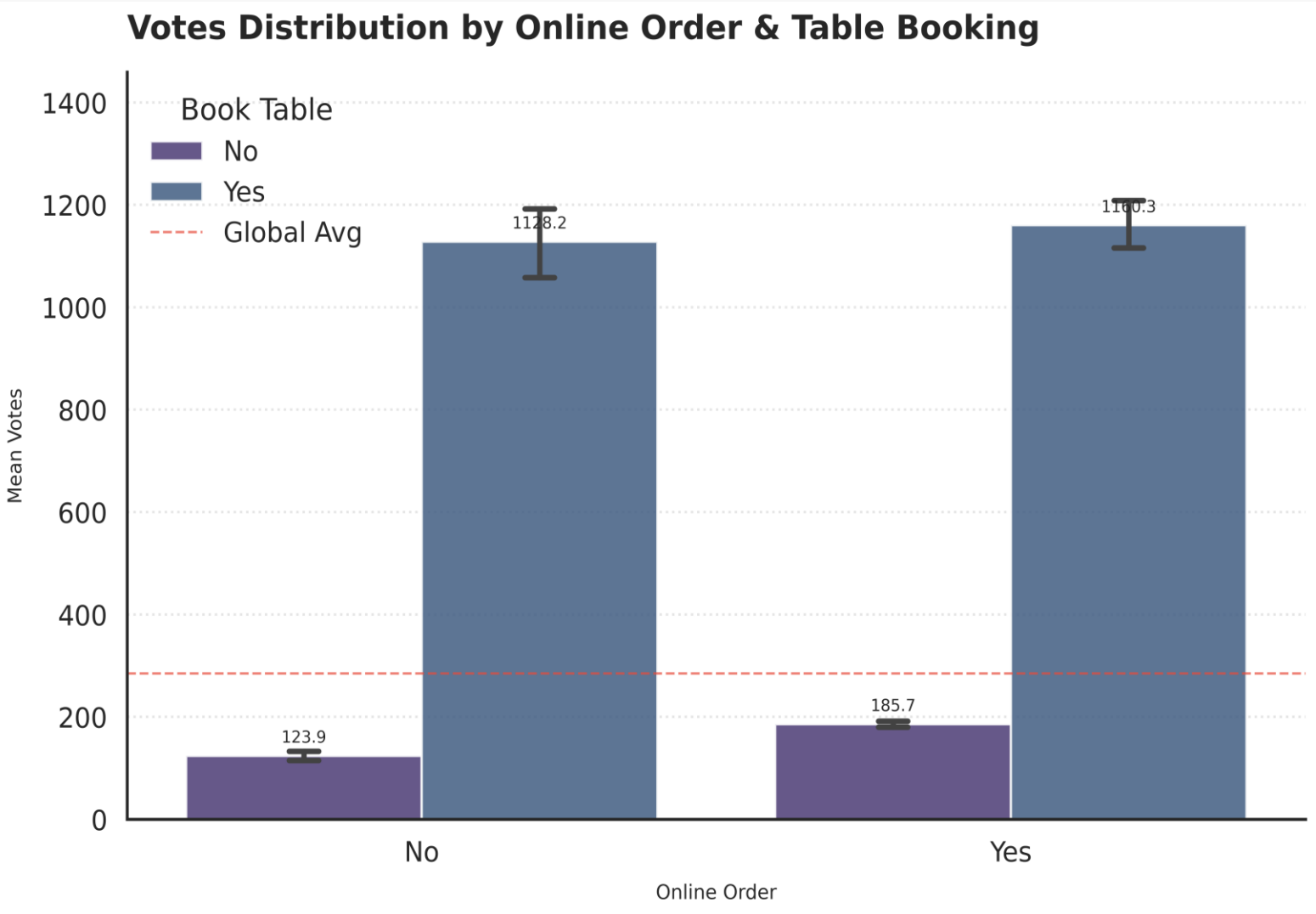
10

category dtype

online_order & book_table — only 2 values each

Result: 51,148 clean records with 0 nulls across all critical columns.

Visualization 1 – Votes by Online Order & Table Booking

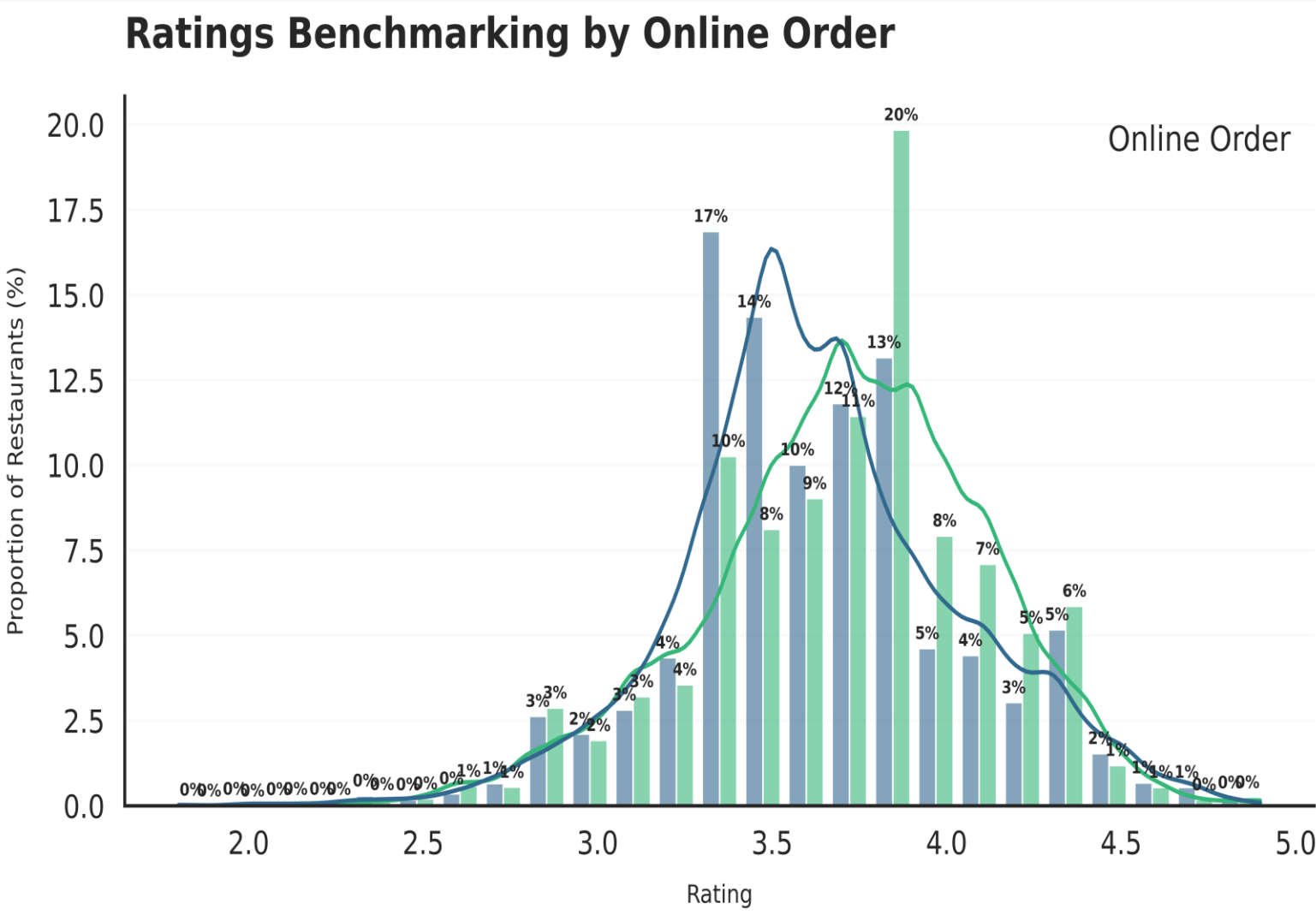


Key Insights

- ▶ **Booking is King:** Table booking drives 7× more engagement than no-booking restaurants.
- ▶ **87.4% miss out:** Most restaurants do NOT offer table booking — huge untapped opportunity.
- ▶ **Online Order alone:** Small boost only (185 vs 124 votes). Table booking is the real lever.
- ▶ **Action:** Enabling table booking is the fastest way to scale visibility on Zomato.

Fig 1. Mean votes segmented by online order & table booking

Visualization 2 – Ratings Benchmarking by Online Order

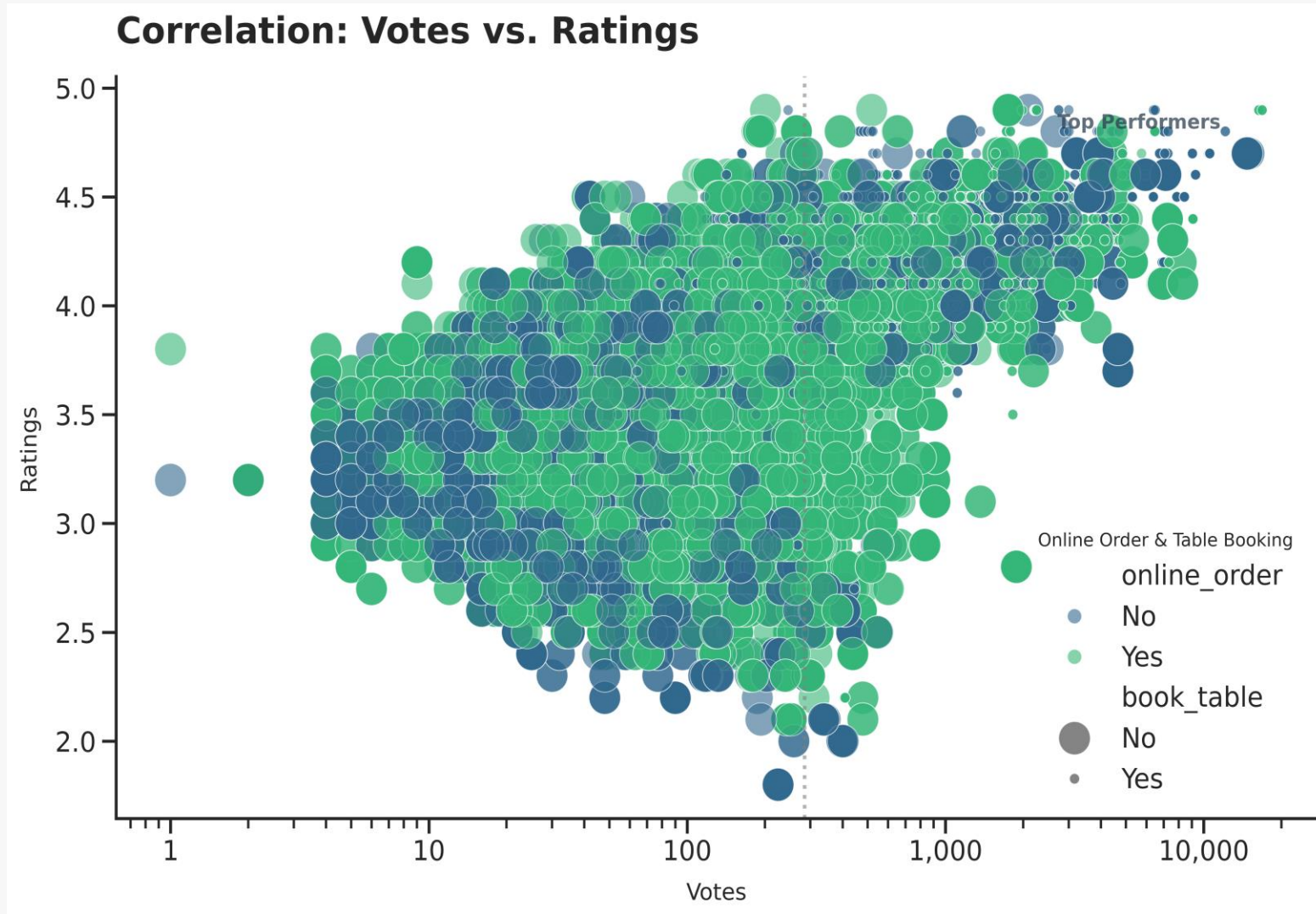


Key Insights

- ▶ **Online = Higher Ratings:** Green curve shifts right — delivery restaurants rate better.
- ▶ **Peak without delivery:** 3.4–3.5 is the modal rating for offline-only restaurants.
- ▶ **Peak with delivery:** Shifts up to 3.7–3.9 — a clear and consistent advantage.
- ▶ **Convenience pays:** Customers reward accessibility with better ratings. Go digital from Day 1.

Fig 2. Proportion of restaurants across rating bands, split by online order

Visualization 3 – Correlation: Votes vs. Ratings

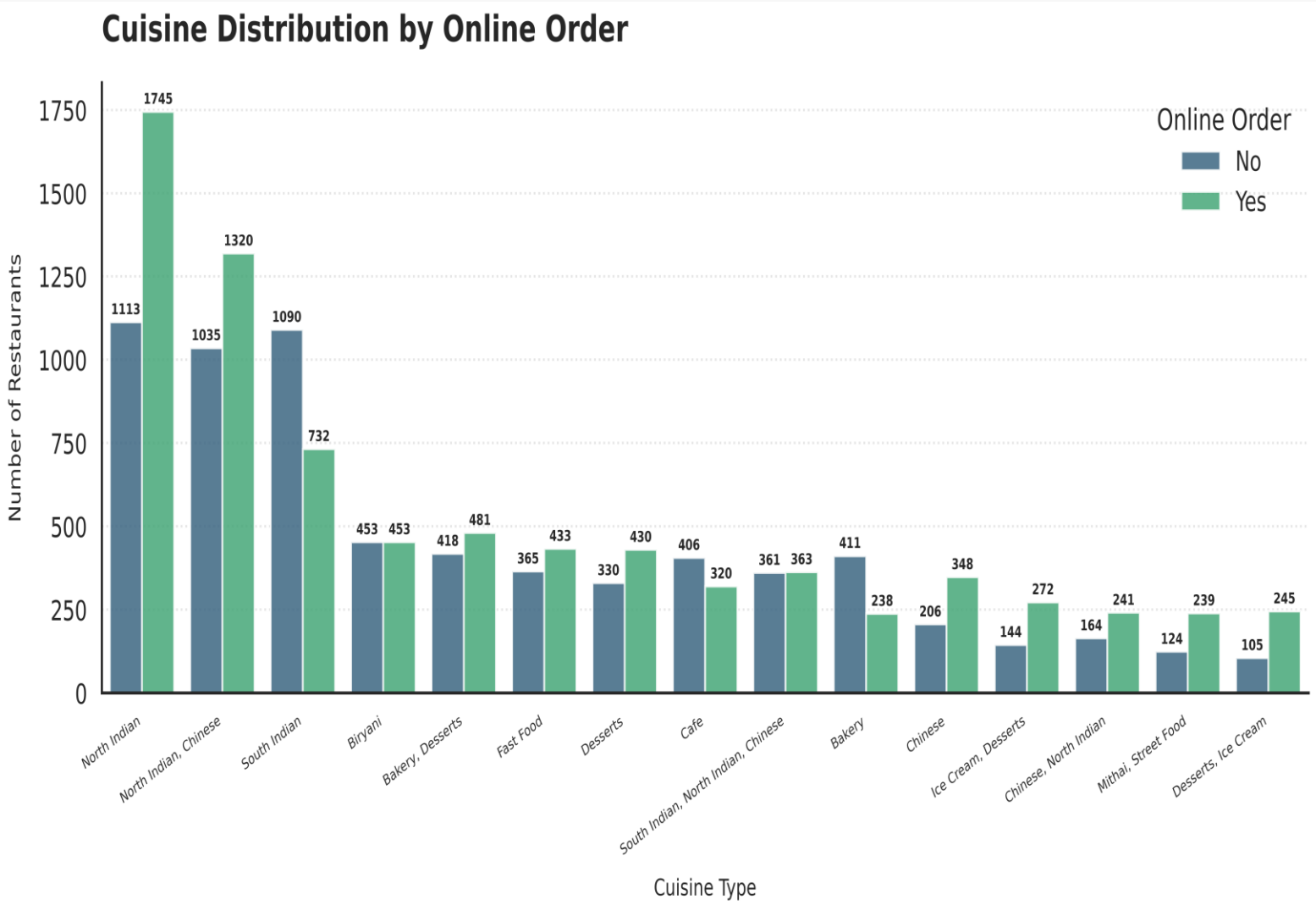


Key Insights

- ▶ **$r = 0.44$** : Moderate positive correlation — more votes generally means higher ratings.
- ▶ **Quality without volume**: Many restaurants (low votes) still score 4.0+.
- ▶ **Market Leader formula**: Top-right — high votes + high ratings — almost all offer both features.
- ▶ **Growth ceiling**: Extremely rare to find 1,000+ vote restaurants without digital features.

Fig 3. Votes vs. Ratings bubble chart — Pearson $r = 0.44$

Visualization 4 – Cuisine Distribution by Online Order

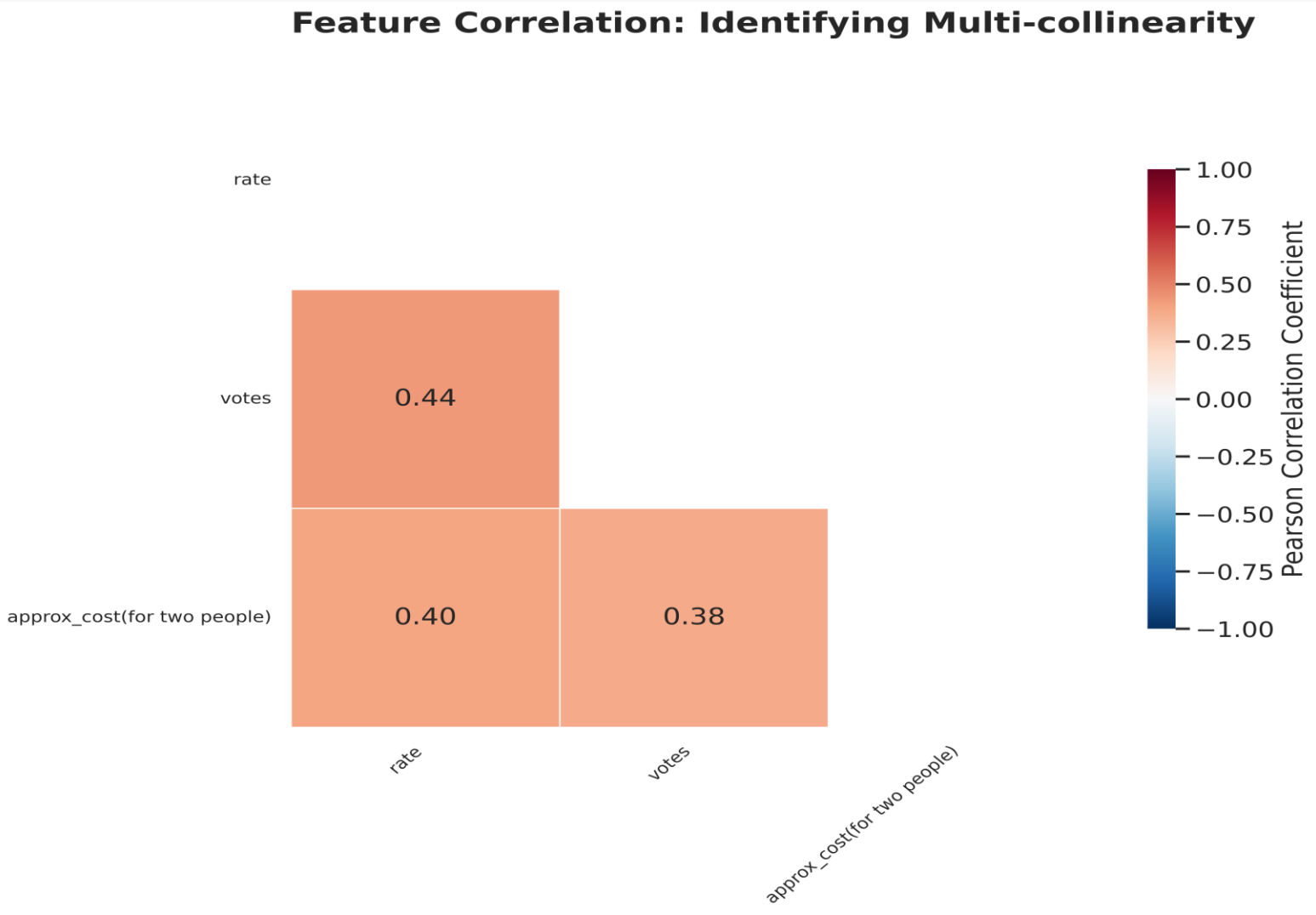


Key Insights

- ▶ **North Indian dominates:** Highest volume but also most competitive segment.
- ▶ **Delivery is standard:** For top cuisines, green bars > blue — digital is table stakes.
- ▶ **Cafe & Bakery gap:** Blue bars nearly equal green — underserved digital audience.
- ▶ **Opportunity:** Cloud kitchen focused on Cafe/Dessert delivery could capture this gap.

Fig 4. Top 15 cuisine types and their online ordering adoption rate

Visualization 5 – Feature Correlation Heatmap



Key Insights

- ▶ **Votes ↔ Ratings (0.44):** Strongest link — quality drives engagement organically.
- ▶ **Cost ↔ Ratings (0.40):** Higher price ≠ guaranteed 5 stars. Quality matters more.
- ▶ **Cost ↔ Votes (0.38):** Premium price does NOT drive traffic alone — service does.

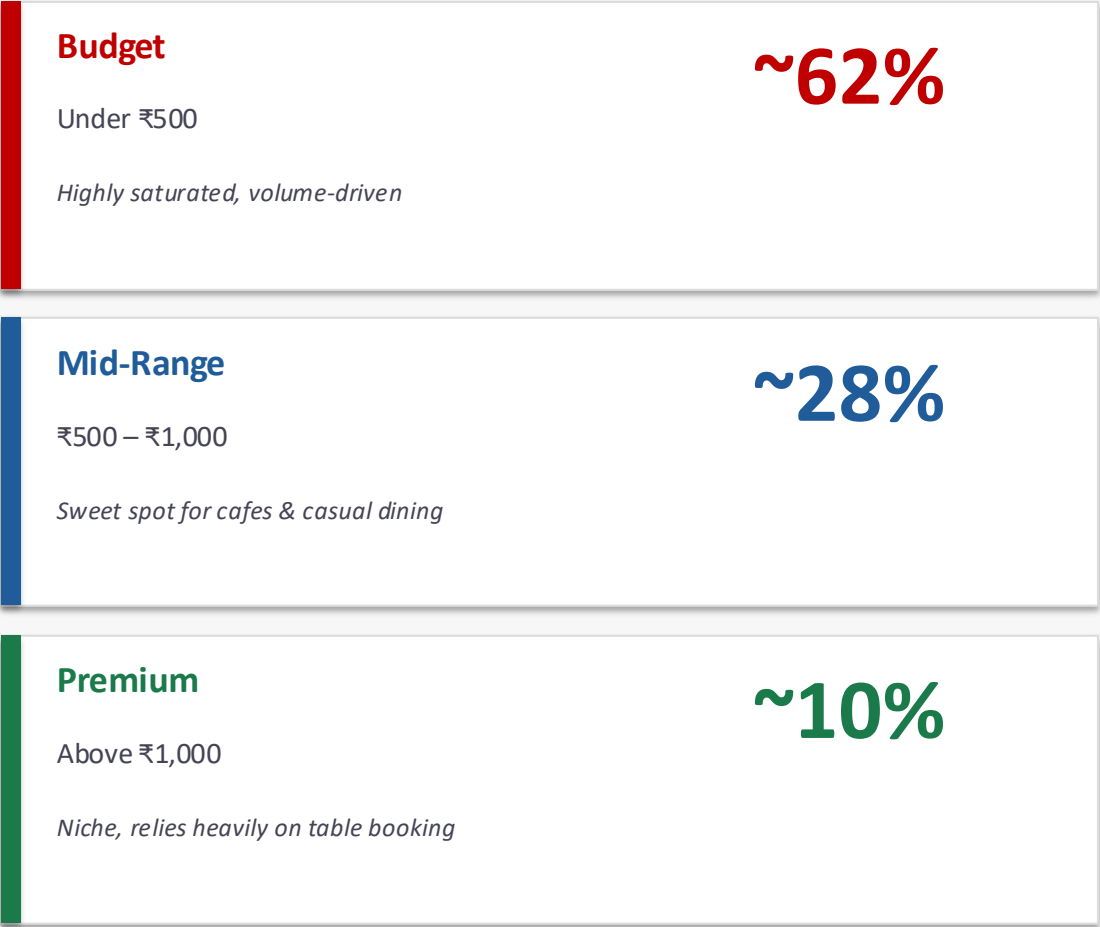
Fig 5. Pearson correlation heatmap — rate, votes, approx_cost(for two people)

Location Analysis & Cost Bracket Segmentation

Top 5 Restaurant Hubs by Volume

Location	Count	Primary Audience
BTM Layout	5,056	Students & Early Professionals
HSR Layout	2,494	Tech Workers & Families
Koramangala 5th	2,479	Mixed — Tech, Students, Expats
JP Nagar	2,218	Families & Residents
Whitefield	2,105	Corporate IT & Expats

Market Share by Price Bracket



Real-World Problems & Data-Backed Solutions

01

The 'Invisible' Restaurant Syndrome

DATA: 19.3% of restaurants (~10,000) have ZERO votes — completely dead digitally.

SOLUTION: Integrate Zomato delivery from Day 1. Walk-ins alone are not enough to build a digital footprint.

02

Extreme Saturation in Budget North Indian

DATA: 62% of market is under ₹500 — dominated by North Indian. Race-to-the-bottom on margins.

SOLUTION: Target the ₹500–₹1,000 'Sweet Spot' or Cafe/Dessert niches with lower online competition.

03

Missing the Biggest Growth Multiplier

DATA: 87.4% of restaurants don't offer table booking — yet it drives 7× more engagement.

SOLUTION: Partner with Zomato Pro, EazyDiner or Dineout. Treat table booking as a marketing expense, not a luxury.

Strategic Recommendations

01 ₹500–₹1,000 Sweet Spot

Avoid the overcrowded Budget North Indian segment. Target mid-range Cafe or Bakery concepts where competition is lower and margins are healthier.

02 Digital-First Launch

Never open without Zomato Delivery + Table Booking integration active from Day 1. These are primary traffic drivers, not optional add-ons.

03 Quality Over Discounts

Consistent 4.0+ ratings organically drive massive engagement ($r = 0.44$). A premium price point is acceptable if service and booking experience are seamless.

04 Table Booking as Marketing Spend

87.4% of restaurants ignore this. Enabling it via Zomato Pro, EazyDiner, or Dineout is the fastest ROI lever available — proven 7× engagement multiplier.

Technical Challenges & Problem Solving

Mobile Dev Environment

Problem: Entire pipeline built and executed on Android using Termux + Acode/Pydroid3. Required strict resource management and efficient coding practices.

Solution: Used PATH variable for cross-environment compatibility; os.path.join() for reliable file handling.

Memory Optimisation

Problem: Handling 51,000+ rows on constrained mobile hardware caused memory bottlenecks.

Solution: Aggressively downcasted dtypes: rate → float16, votes → int32, categoricals → category dtype. Significant RAM reduction.

Inconsistent Data Formats

Problem: Ratings stored as '4.2/5', 'NEW', or '-'. Cost had comma separators ('1,000').

Solution: Used Pandas string manipulation (.split('/'), str.replace(',', '')) + pd.to_numeric(errors='coerce') for robust parsing.

Smart Missing Value Imputation

Problem: 7,775 missing ratings — naive global average would distort location/type-specific patterns.

Solution: groupby(['rest_type', 'location'])['rate'].transform('median') — preserves local context. Global median as fallback.

Conclusion

This EDA of 51,148 Zomato Bangalore restaurants transforms raw, messy data into actionable business intelligence. Success in this highly competitive market is driven less by operating in overcrowded budget segments, and more by digital presence — specifically Table Booking and Online Delivery integration.

Executed entirely within a constrained mobile development environment (Termux + Acode) using Python and Pandas, this project demonstrates end-to-end analytical capability — bridging raw data manipulation with high-level strategic consulting.

Data Disclaimer

In real-world data science, 100% data purity is a myth. Claiming a 'perfectly clean' dataset often signals that data was forcefully sanitized — blindly dropping nulls or outliers without understanding their business context.

This analysis preserved data integrity through smart imputation (groupby median, not blind deletion), retaining maximum business value. The insights and recommendations therefore operate within a realistic and highly reliable 90–95% confidence range.